

STS-Net: A Phonological Property Prediction Model for Swedish Sign Language

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Abstract

We present STS-Net, a neural model that predicts phonological properties of Swedish Sign Language (SSL) signs directly from 2D pose keypoints. Given a short isolated-sign video clip, the model predicts handshape, hand attitude, contact location and type, motion direction, and hand type via an attention-pooled multi-stream encoder trained on the Swedish Sign Language Lexicon (SSLL). We show that the learned embeddings can be transferred to a large continuous-signing corpus (SSLC) through nearest-neighbour mining, yielding a 5-percentage-point gain in handshape accuracy. The enriched representations improve gloss retrieval by a factor of six and support competitive sign boundary segmentation on SSLC continuous signing.

Index Terms: sign language, phonology, keypoint estimation, sequence modelling, segmentation

1. Introduction

Sign languages have a rich sublexical phonological structure: each sign is characterised by handshape, hand orientation (attitude), place of articulation, contact type, movement, and number of hands. Computational modelling of these properties supports recognition, segmentation, and linguistic analysis. Tavella et al. [1] introduced WLASL-LEX, a dataset for recognising phonological properties in ASL from skeleton features, demonstrating the feasibility of automatic phonological annotation from pose. Labelled phonological data remain scarce for smaller sign languages, however, and continuous-signing corpora are typically unannotated at the phonological level.

Swedish Sign Language (STS) is one such language. The Swedish Sign Language Lexicon (SSLL)¹ provides roughly 20,000 isolated-sign video clips, each accompanied by a textual description from which phonological targets can be automatically parsed. The Swedish Sign Language Corpus (SSLC), by contrast, contains approximately 210,000 annotated gloss instances in continuous conversational signing from 37 signers, but carries no phonological labels [2].

This paper presents STS-Net, which learns six phonological feature categories from SSLL isolated clips and demonstrates three downstream applications: (1) mining SSLC for additional training data via embedding-space retrieval, (2) gloss recognition in SSLC using the learned representations, and (3) sign-boundary segmentation of SSLC continuous video.

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2. Model

2.1. Input representation

Each clip is processed with MediaPipe [3] at 25 fps to yield four parallel pose streams: (i) dominant hand — 21 wrist-centred keypoints; (ii) non-dominant hand — 21 keypoints; (iii) upper body — 12 shoulder-normalised keypoints; (iv) face — 25 nose-normalised keypoints. All keypoints are 2D (x, y); invalid frames (missing wrist detection) are zeroed.

2.2. Architecture

Each stream is encoded by a dedicated **FrameEncoder**: a linear projection to $D=256$ dimensions, layer normalisation, ReLU activation, and three stacked temporal convolution blocks (Conv1d, kernel 5, residual connections, layer norm, ReLU, dropout). The four 256-dim outputs are concatenated and passed through a **Fusion MLP** ($1024 \rightarrow 256$, layer norm, ReLU, dropout). A **masked attention-pooling** layer then aggregates frames within the annotated sign window [start, end] into a single 256-dim clip embedding.

Six multi-label **classification heads** read from this embedding: handshape (42 classes, multi-hot BCE), attitude/orientation (34 classes, BCE), contact location (22 classes, BCE), contact type (4 classes, BCE), motion direction (8 classes including *none*, BCE), and hand type (2 classes, cross-entropy). Total parameters: $\approx 4.3\text{M}$.

2.3. Training

The model is trained on $\approx 17,700$ SSLL clips (signer-based 85/15 train/val split) with AdamW ($\text{lr}=3\text{e-}4$), batch size 32, cosine LR schedule, dropout 0.2, and Gaussian pose noise ($\sigma=0.005$). Phonological targets are parsed automatically from the textual sign descriptions. Base validation accuracy: handshape 80.1%, attitude 77.4%, hand type 96.3%.

3. Experiments

3.1. Input representations

Table 1 compares five input representations on SSLL. **MediaPipe (MP)** is the 2D skeleton baseline. **WiLoR** replaces hand streams with 3D MANO keypoints [4]. **WiLoR+MP** combines all six streams. **DINO+MP** adds a DINOv3 ViT-L/16 CLS token [5] to the skeleton. **Sapiens** uses a whole-body pose foundation-model stream in place of MediaPipe.

WiLoR+MP scores best across all properties, confirming 3D hand geometry complements the 2D skeleton. Sapiens matches the MP baseline despite using a single stream. DINO lags, as CLS tokens capture global appearance rather than finger articulation.

Table 1: *Top-1 accuracy (%) by input representation (SSL only, val set).*

Property	MP	WiLoR	WiLoR+MP	DINO+MP	Sap.
Shape (dom.)	85.1	83.5	85.5	82.5	83.0
Attitude	78.8	79.3	79.3	76.7	78.4
Contact loc.	81.2	79.7	81.8	77.7	81.1
Contact type	75.8	75.7	76.3	73.3	76.1
Motion dir.	63.6	63.1	65.1	60.5	64.4
Hand type	97.0	96.7	97.3	96.5	96.7

3.2. Sign mining from continuous data

Phonological labels cannot be derived directly from SSLC gloss strings. Instead, we transfer labels via embedding-space matching. For each SSL training clip we compute a 256-dim clip embedding; embeddings are grouped by gloss. For each SSLC gloss window we compute the same embedding and retrieve the nearest SSL variant per gloss. Instances with cosine distance below 0.4 are retained, and the SSL clip’s phonological targets are transferred.

This yields **11,280 instances** spanning 1,284 glosses from 28 SSLC signers. Re-training on SSL + mined SSLC with stronger regularisation (dropout 0.3, noise $\sigma = 0.02$, label smoothing 0.1, $\pm 15\%$ temporal stretch) gives the improvements in Table 2.

Table 2: *Top-1 accuracy on all predicted phonological properties before and after SSLC mining. Both models use 2D pose, 4 streams, and a non-dominant handshape head. Evaluated on the SSL validation set (3,100 clips).*

Property	SSL only	+mined SSLC
Handshape (dom.)	79.9%	85.1%
Attitude (dom.)	75.9%	79.3%
Contact location	78.3%	80.6%
Contact type	72.3%	76.8%
Motion direction	57.4%	62.4%
Hand type	96.6%	97.1%
Handshape (nondom.)	81.3%	85.7%

3.3. Gloss recognition on SSLC

Zero-shot retrieval. Each of 64,462 SSLC test clips is ranked against per-gloss mean SSL embeddings over 2,010 glosses. Mining raises Rec@1 from 1.0% to **5.8%** and Rec@10 from 5.8% to **22.2%** without any direct SSLC supervision for this task.

Supervised linear probe. A linear classifier is trained on frozen 256-dim clip embeddings from 116k SSLC clips and tested on 13k clips (1,824-gloss closed vocabulary), as shown in Table 3. The mining-augmented model outperforms the SSL-only baseline by ≈ 4 percentage points, with median rank improving from 12 to 8.

Table 3: *Supervised linear-probe gloss recognition on SSLC.*

Model	Rec@1	Rec@5	Med. rank
SSL only	18.6%	38.3%	12 / 1824
+mined SSLC	22.8%	44.5%	8 / 1824

3.4. Sign segmentation on SSLC

We address BIO sequence labelling (B=begin, I=inside, O=outside) for sign and phrase boundaries on SSLC continuous video, evaluated on 44 held-out test videos (signer-based split). A **TCN segmentation head** (6 dilation levels, kernel 5, receptive field ≈ 10 s at 25 fps) is trained on top of frozen STS-Net frame features extracted per video. Table 4 compares this against a windowed linear classifier, a raw-pose BiLSTM replicating the architecture of Moryossef et al. [?], and their pre-trained model applied zero-shot from German Sign Language (DGS).

Table 4: *Sign/phrase segmentation on SSLC (44 test videos). Sign % = predicted / gold segment count (ideal: 1.00). The DGS in-domain row is not directly comparable.*

Model	S-F1	S-IoU	P-F1	P-IoU
Windowed linear	0.540	0.440	0.486	0.372
BiLSTM raw pose	0.532	0.531	0.527	0.582
Pretrained DGS→SSLC	0.498	0.442	0.569	0.612
TCN on STS-Net (ours)	0.547	0.511	0.535	0.572
<i>Moryossef ’23 (DGS)</i>	<i>0.630</i>	<i>0.690</i>	<i>0.650</i>	<i>0.820</i>

Our TCN achieves the best sign-boundary F1 among SSLC-trained models (S-F1=0.547) and produces near-ideal sign count ratios (Sign % ≈ 1.00). The pretrained DGS model leads on phrase IoU in zero-shot transfer — phrase-level structure appears more cross-lingual than sign identity. The gap to the in-domain DGS reference suggests that SSL-native continuous training data or cross-lingual transfer would yield further gains.

4. Conclusions

STS-Net is a compact, pose-based model that predicts six phonological feature categories from short SSL clips with competitive accuracy. A nearest-neighbour mining step bridges isolated and continuous signing corpora, yielding a 5 pp gain in handshape accuracy and a sixfold improvement in zero-shot gloss retrieval. The frozen backbone supports competitive sign boundary segmentation on SSLC continuous video. The model is available at github.com/jbeskow/stsnet

5. References

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